

Matthew Molineaux, Ph.D.

AI SYSTEMS ARCHITECT · AUTONOMY & AI AGENTS

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PROFILE

AI systems architect, seeking architecture and technical leadership in robotics, autonomy and medicine. Twenty-three years architecting and building autonomous agents and AI-assisted reasoning tools. Philosophy: language models for open-ended language work; explicit, transparent models wherever precision matters. Recent work: agent for website evaluation with simulated customers for under \$5 a run (solo architect and developer), and a medical triage decision system that beat the generative-AI competitor in all four independent DARPA evaluations (PI and lead developer).

Seventeen AI systems architected, with evidence: molinemc.github.io/systems.html

TECHNICAL FOCUS

Autonomous agents — Agent architectures; goal reasoning and goal formulation; hierarchical and temporal planning; causal explanation and diagnosis; execution monitoring and plan repair; human-agent teaming; constraint and combinatorial optimization; answer set programming; computer-use agents; general AI architectures portable to new problems with declarative domain knowledge

Learning and evaluation — Reinforcement learning; environment and action model learning; experiment design and metric definition; independent and adversarial evaluation; human-alignment measurement; open-world robustness under distribution shift

EXPERIENCE

Independent AI Product Development

Feb 2026 – Present

Founder

Washington Township, OH

Telotrope Agents, Inc. (Jun 2026–Present) · Delegation, Inc., co-founder (Feb–Aug 2026)

- Built a website-evaluation tool and agent solo, with LLM-assisted implementation, automated tests and a Docker deployment. Agent walks a website as one to twenty diverse customers with distinct goals, budgets and expertise. Tool reports each persona's path, questions the website answered and failed to answer, conversion decisions, pages no visitor found. Tested across fifty sites; three retail prospects received reports and registered interest.
- Held every run under \$5: a representative 13-site batch cost \$0.85–\$2.96 per site in 13–26 minutes. Each persona asks the hard questions a site's designers may never have anticipated, and the site is judged on how many it answers: 53–92% across that batch. Tool reports where anti-bot defenses and non-HTML widgets limited the agent's reach.
- Briefed two DARPA program managers, submitted an RFI response to SCO and an ERIS proposal, and developed teaming relationships with Kairos, Knexus and NRL.

Parallax Advanced Research

Feb 2018 – Apr 2026

Director of AI and Autonomy

Beavercreek, OH

- Principal Investigator and lead developer on a \$4.1M DARPA award.¹ Architected the Trustworthy Algorithmic Delegate (TAD), which performs decision-maker alignment: deciding how to triage with internalized judgment from particular doctors. Uses case-based reasoning for sample-efficient learning, Monte Carlo simulation for prognosis evaluation, Bayesian diagnosis for recognizing symptom causes, and RAG-based question answering for suggesting

treatments to consider internally. Directed ten contributors across four institutions.²

- TAD answered in under a second on local hardware, deployed in Docker, across REST/JSON interfaces defined jointly with BBN and SoarTech. Met or exceeded targets in all four independent evaluations run by CACI, on schedule, ahead of the competing generative-AI team every round, with no decision left to generation. The evaluation team reported “impressive progress”,³ and DARPA featured the work in its Voices from DARPA podcast.⁴
- Co-organized HAADMS, the workshop on Human Alignment in AI Decision-Making Systems at IEEE CAI 2025, in support of DARPA's ITM program; DARPA links its proceedings.⁵ Spoke on goal reasoning and dynamic replanning for uncrewed systems in an AUVSI industry webinar.⁶
- Built and ran a 10-person group across AI, machine learning, neuromorphic computing and cognitive science, and directed AFRL projects with staff matrixed to multiple customers. Hired about 10 technical staff; three now lead their own projects, two were promoted. Graduated 3 PhD and 2 MS students.
- Owned five winning bids worth more than \$15M, including In the Moment (ITM) and PRIDE, from problem framing through architecture, pricing and staffing, with repeat awards from AFRL and DARPA. Originated multiple bids, including ITM and PRIDE, finding the opportunity and building the team.
- Architected three further systems: GATE, which hands a team in crisis a small change to its plan rather than an entire new plan, preserving dependencies between teammates' tasks (teams responded significantly faster to both complications, and were significantly more effective); PRIDE, a theory-of-mind decision aid held to a laptop-sized footprint⁷; and a hierarchical goal planner for the \$4.4M DARPA SCEPTER program, integrated with the host system over MQTT, and built with two Parallax and three SIFT scientists.
- Created GASR, an original planner that splits planning tasks at delegated goals, cutting search from hundreds of nodes to about forty. Rebuilt it in Rust in 2026.
- Architected a framework for measuring how agents cope with change: it transforms an environment deliberately, along eight dimensions of novelty, and measures how far performance degrades.

Knexus Research Corporation

Senior Computer Scientist

Apr 2007 – Jan 2018

National Harbor, MD

- Created ARTUE (Autonomous Response to Unexpected Events), a domain-independent agent that responds to surprise by explaining the surprise's cause and dynamically updating its own goals. Explaining surprises and adjusting goals improved agent adaptation: in one scenario, replanning alone reached 48% performance, while ARTUE's surprise response reached 98%. ARTUE's planner, SHOP2-PDDL+, included a first-of-its-kind ability to plan based on non-linear continuous effects, with hierarchical decomposition. Produces 50-action plans in about a second.
- Invented DiscoverHistory, an algorithm that reasons backward from partial observations to unobserved events. As accurate as a state-of-the-art deductive method, requiring 5% of the time; an agent using it achieved 20–63% more goals than one that does not explain. EML, a learner working in tandem with it, repairs the agent's world model using event explanations, reaching 90% of the achievable improvement within five trials.
- Built three agents on the TIELT and LIET integration layers, each learning something a controller cannot be told in advance: CaT, a case-based system that learns to defeat a real-time strategy opponent it has never seen; CASSL, which learns continuous action models where discretization loses the order the situation calls for; and Casey, a transfer learning system that recognizes an adversary's plan and reuses that recognition to control its own side, learning ten times faster than competitors without it. A video of Casey won Best Overall at IJCAI 2009.^{8,9}
- Architected replanning-and-execution systems for physical operations: CARPE and its second

generation, AMORE, plan and schedule weeks-long mine countermeasures operations with dozens of units; ADRUS detects and resolves anomalies in unmanned logistics operations. CARPE's scheduler, CLOSR, repaired plans with minimal disruption and still met the mission standard across randomly generated operations; the Navy funded a second generation. Contributed significantly to six winning proposals, worth \$6.5M, that funded this and related work.

- Built ASM, an agent that joins a human squad without asking anything of them: it infers what the team is doing from observation and speech they would utter anyway, then picks its own goal to match. Demonstrated end to end: a team interrupted by a sniper abandons the mission goal, clears the threat and resumes, with the agent tracking all of it from observation alone.

ITT Industries

Dec 2003 – Apr 2007

Research Scientist

Alexandria, VA

Full-time under contract to the Naval Research Laboratory, Washington, DC

- Architected and built TIELT, the integration layer for a DARPA program, turning an $M \times N$ interface problem into $M+N$: one interface per decision system and one per environment rather than one per pairing. Seven environments behind it, from Unreal Tournament to Chess to Zork to Gazebo/Stage, using customizable interfaces to each. A Georgia Tech team used it for DARPA transfer-learning work with a commercial game engine from Mad Doc Software (MadRTS).
- Helped design the metrics for the DARPA Transfer Learning program's technology evaluations, and analyzed results across performer teams.

EDUCATION

Ph.D., Computer Science, George Mason University	2017
M.S., Computer Science, DePaul University	2008
B.S., Computer Science, Eckerd College	2001

RECOGNITION AND PUBLICATIONS

DARPA Riser, 2022 ¹⁰ · Best Paper: ICCBR 2005, FLAIRS 2008 and 2011 · Best Overall Video, IJCAI 2009 (from 39 entries)

65+ publications in planning, goal reasoning, explanation and agent evaluation.

REFERENCES

1. Parallax Advanced Research wins DARPA In the Moment award totaling \$4.067M, 2023
2. Drexel University joins the DARPA In the Moment effort, 2023
3. McVay, J. and de Visser, E. Challenges and Choices when Evaluating Alignment in Human-AI Systems, AAI Fall Symposium 2025
4. DARPA. When Should Machines Decide? Voices from DARPA, ep. 83, 2024
5. HAADMS 2025, IEEE CAI: papers and presentations
6. AUVSI webinar. Advancing Mission-Level Autonomy: Dynamic Replanning, Goal Reasoning, and the Future of Human-Machine Teaming, 2026
7. Knexus Research Corporation and Parallax Advanced Research advance DARPA's PRIDE Project
8. Casey's Quest: Transfer Learning for Adversarial Environments (IJCAI 2009 Best Overall Video)
9. Naval Research Laboratory. NRL artificial intelligence team win prestigious video awards, 2009
10. Parallax Advanced Research scientist selected for DARPA Riser Program, 2022

Tools — C++, Rust, Python, Common Lisp · PDDL, Clingo/ASP · Claude, open-weight LLMs, PyTorch, MCP, Playwright · Docker, GitLab CI, pytest, JUnit, Gradle · MQTT, AWS, ROS 2